

Physics Puzzle Platformer

2D platformer built on a custom verlet soft-body solver, with full gamepad support, Steam achievements, and 40 hand-authored levels.

Unity	C#	Shader Graph	Steamworks
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VERSION	STATUS	LEVELS	ENGINE	PLATFORMS	UPDATED
1.4.2	Live	40	Unity 6	PC, Deck	Sep 2026

1. Overview

A 2D physics platformer whose central mechanic is deformation. The player character and most puzzle objects are soft bodies simulated with a verlet particle solver and distance constraints rather than rigid colliders, so squeezing through a gap, absorbing a fall, or stretching across a chasm are all emergent properties of the same simulation.

Unity's built-in 2D physics is not used for anything that matters. The solver runs on a fixed timestep with its own broadphase and collision response, because soft-body stability depends on controlling substep count and constraint iteration order — neither of which an off-the-shelf rigid-body engine exposes. Unity supplies rendering, input, audio, and packaging.

2. Simulation frame

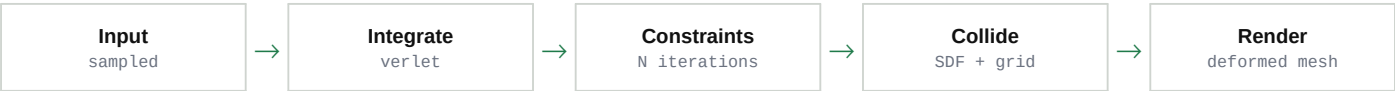


Figure 1 — Four substeps per 60 Hz frame. Constraint iterations run inside each substep, not once per frame.

Level geometry is baked into a signed distance field at load, which makes collision a field lookup and a gradient rather than a shape intersection test. That is what keeps a 400-particle soft body affordable: each particle resolves against the world in constant time regardless of how complex the level's silhouette is.

3. Systems

Solver	Verlet integration with distance, bend, and volume-preservation constraints, four substeps.
Collision	Baked SDF for static geometry; uniform grid broadphase for particle-particle contacts.
Character	Input mapped to internal pressure and anchor impulses rather than to a velocity vector.
Level format	Authored in Unity scenes, exported to a compact binary with the SDF baked in.
Platform layer	Steamworks achievements and cloud saves, full gamepad and Steam Deck input maps.
Telemetry	Opt-in level completion and death-position events used for difficulty tuning.

4. Performance

PLATFORM	FRAME TIME	NOTES
Desktop, mid-range GPU	4.1 ms	Solver 2.3 ms of that; 1080p, vsync at 60.
Steam Deck	12.8 ms	Holds 60 Hz; solver substeps unchanged from desktop.
Worst-case level	16.2 ms	Chapter 5 chain puzzle, nine simultaneous soft bodies.

5. Level definition

```
{
  "id": "c3_07_pressure",
  "chapter": 3, "par_time": 74.0,
  "sdf": { "source": "c3_07.geo", "resolution": 0.05, "bounds": [64, 36] },
  "spawn": { "x": 4.0, "y": 6.5, "material": "default_blob" },
  "objects": [
    { "type": "soft", "mesh": "crate_soft", "stiffness": 0.62, "mass": 4.0 },
    { "type": "hinge", "anchor": [22.0, 14.0], "arm": 3.5, "damping": 0.2 },
    { "type": "field", "kind": "updraft", "rect": [30, 2, 6, 18], "force": 9.0 }
  ],
  "goal": { "type": "volume_in_zone", "rect": [58, 4, 5, 5], "fraction": 0.8 },
  "achievements": ["c3_no_deaths", "c3_under_par"]
}
```

Listing 1 — Goals are stated as solver conditions, so a level is won by the simulation rather than by a trigger volume.

6. Solver stability

- **Substepping over iteration.** Four substeps with eight constraint iterations each proved far more stable than one substep with thirty-two, particularly for fast impacts, at almost identical cost.
- **Constraint ordering.** Distance constraints are solved in a fixed, mesh-authored order rather than by index, which removes the directional bias that made early builds drift left on landing.
- **Clamping.** Per-particle velocity and constraint correction are both clamped. A soft body can be crushed to zero volume without exploding, which matters because players will try exactly that.

7. Shipping and operations

- Builds are produced by a headless Unity job that runs the level validation suite first: every level is solved by a recorded input replay, and a level that no longer completes blocks the build.
- Achievements are defined in the level data and registered with Steamworks at build time, so a renamed achievement cannot silently detach from its trigger.
- Death-position telemetry is aggregated into per-level heat maps, which is how difficulty spikes in chapters 4 and 5 were identified and retuned after launch.

8. Roadmap

RELEASE	PLANNED WORK	TARGET
1.5	Chapter 6: twelve levels introducing temperature-driven stiffness.	Q4 2026
1.6	Level editor and Steam Workshop publishing for community levels.	Q1 2027
2.0	Console ports and a two-player cooperative mode sharing one solver.	Q3 2027